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Acoustic UAV detection method based on blind source separation framework

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ABSTRACT

Many new challenges are encountered in public security because of the rapid development of an unmanned aerial vehicle (UAV) technology. In order to meet these challenges brought by UAV, it is necessary to realize timely and accurate UAV detection. In this paper, an acoustic UAV detection method based on blind source separation (BSS) framework is introduced to solve the UAV sound detection problem with multi-source interference. In the framework proposed in this paper, source number estimation is used to determine the type of BSS problem, and three methods are applied to separate the UAV sound from the mixing signal in different BSS situations. In the experiment, a variety of machine learning algorithms are used to detect UAVs. The robustness of the proposed method is tested on different UAVs and different sound features. Experimental results show that the UAV sound detection method based on the BSS framework realizes the effective detection of UAVs in different tests and achieves excellent robustness. Compared with mixed signal and filtered signal, the detection rate of our scheme is more than 90%, whether in the overdetermined, positive-definite, and underdetermined cases.

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1. Introduction

UAVs have shown great potential in the fields of mobile communications, Internet of Things (IoT), agriculture, environmental monitoring, urban transportation and disaster relief. UAVs are applied in predicting crop yield [1], estimating wheat plant density [2], and monitoring aboveground biomass over large areas in agriculture [3]. The application of UAVs in environmental monitoring provides great convenience in field surveying and mapping [4]. UAVs are promising enablers of the IoT vision because of their agile maneuverability [5], and they are gradually becoming a key part of any smart city concept [6,7]. Meanwhile, the accurate assessment of building damage with UAV has become a research hotspot [8,9]. A recent study showed that a UAV team can monitor a wildfire in open space closely and track its development precisely [10]. However, the overuse of UAVs has also caused many problems in security and privacy. As shown in Fig. 1(a), UAVs can be easily used by criminals for drug transportation because of their simple operation and cost-effectiveness. Terrorist attacks using UAVs have also been frequently reported (Fig. 1(b)). Private UAVs have penetrated airports, government agencies, and other public places because of the lack of management for such vehicles (Fig. 1(c) and (d)). Undoubtedly, these issues pose new challenges to public safety.

Therefore, the study on anti-UAV technology has attracted increasing attention from many scholars. There are some difficulties when traditional radar technology is used to detect multi-rotor UAV due to the low altitude, small size and slow speed characteristics of UAV. In order to deal with these situations, many new UAV detection methods such as based on image, sound and radio frequency (RF) technology have been proposed.

UAV detection using images has undergone great progress owing to the development of image detection. BorHorng et al. [11] designed a dual-axis mechanism with a thermal imaging camera and full color camera for tracking and locking UAV. Liu Yicheng et al. [12] introduced a UAV detection system by capturing images of flying objects and their trajectories. However, the image-based UAV detection scheme is not applicable in poor weather conditions.

RF technology also offers wide applications in the field of UAV detection. Fu Hua et al. designed a prototype of portable general software radio peripheral equipment, which detects UAVs through the micro Doppler characteristics of RF signals or the

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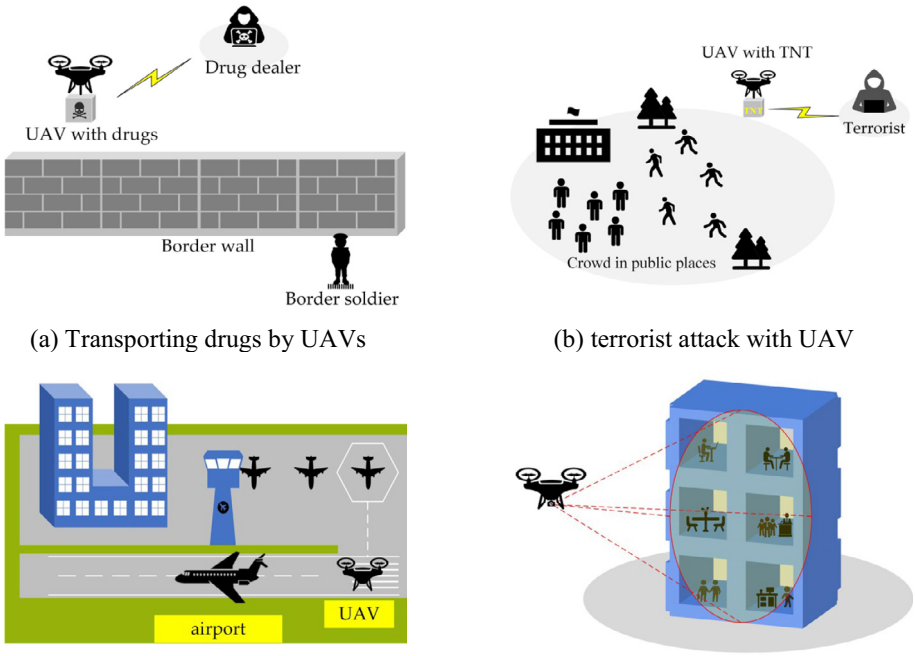


(a) Transporting drugs by UAVs (b) terrorist attack with UAV

(c) Private UAV invades Airport (d) UAV impairs public privacy

Fig. 1. Threats and challenges of UAVs.

cyclostationarity characteristics of UAV signals [13]. Yang Shengying et al. proposed a UAV detection scheme on the basis of an improved RF algorithm to reduce noise and improve signal-to-noise ratio [14]. However, the RF-based UAV detection scheme is often constrained by a complex electromagnetic environment and high equipment cost.

Sound-based UAV detection is a low-cost solution. Blanchard et al. used a microphone array and real-time signal processing algorithm to detect UAV on the basis of the sound emitted by UAV motorization [15]. Muhammad Zohaib et al. introduced a UAV recognition scheme based on machine learning framework, which uses support vector machines with different kernel functions to detect UAV [16]. Projection pursuit can be applied successfully to separate UAV sounds from other various unvoiced sounds [17]. Independent component analysis (ICA) is proposed to solve the UAV sound detection problem in a strong interference environment [18]. The scheme in [12] realizes UAV detection on the basis of raw UAV sounds, but it does not consider scenes with strong interference. The schemes in [17] and [18] are focused on the problem of UAV detection with interference scenarios, but they are only applicable to cases in which the number of sound sources is equal to the number of sensors (positive-definite). Therefore, a UAV sound detection scheme suitable for multiple scenarios (overdetermined, positive-definite, or underdetermined) is urgently needed. Accordingly, we propose an acoustic UAV detection scheme based on the BSS framework that can realize UAV detection in overdetermined, positive-definite, and underdetermined cases with multiple interference sources. The proposed scheme integrates advanced acoustic processing and BSS technologies to enhance the UAV detection performance. This paper has two main contributions:

(1) A comprehensive UAV sound detection solution is introduced in overdetermined, positive-definite, and underdetermined cases.
(2) Source number estimation is used to avoid presupposing the relationship between the number of sound sources and sensors.

The remainder of this paper is organized as follows. The details of the acoustic UAV detection scheme based on the BSS framework are discussed in Section 2. The experiment based on MATLAB is detailed in Section 3. Finally, conclusions are drawn in Section 4.

2. Methods

In the acoustic anti-UAV scheme, the UAV detection rate depends on the quality of UAV acoustic signal. However, obtaining a pure UAV sound signal is difficult because of environmental noise and interference. In this paper, BSS theory is utilized to reduce the interference and noise in UAV sound signals and thereby greatly improve the performance of UAV detection. Source number estimation is the first step in the UAV detection method based on the BSS framework. If the number of sources is obtained, then the type of problem (overdetermined, positive-definite, or underdetermined) can be confirmed. According to problem types, different algorithms are used to separate UAV sounds from the signal containing interference.

A fast-fixed-point algorithm for ICA (FastICA) [19–21] with negative entropy maximization is used to separate the source signal in the positive-definite case. Principal component analysis (PCA) is adopted to reduce the dimension of signals in the overdetermined case. In the underdetermined case, variational mode decomposition (VMD) is used to construct virtual channels. Finally, UAV detection is performed using different machine learning algorithms.

The work flow of our scheme is shown in Fig. 2.

2.1. Blind source separation theory

BSS is a traditional and challenging problem in signal processing and is considered a "cocktail party problem" [22,23]. In a typical cocktail party scenario, the voice from a group of people talking in a room is recorded by a group of microphones. The task of BSS is to separate the target voice signals from the signals recorded by the microphones.

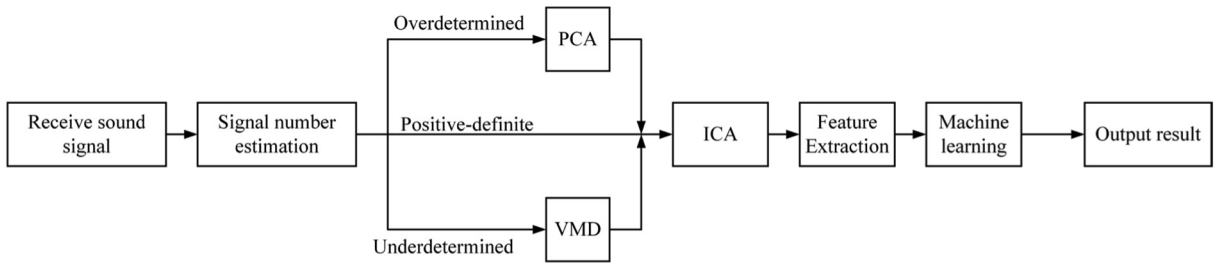


Fig. 2. UAV detection system work flow.

To clearly state and simplify the cocktail problem, several necessary assumptions are listed as follows:

(1) m source signals $S = (s_1, s_2, \ldots, s_m)^T$ exist.
(2) The sensors receive n observed signals $X = (x_1, x_2, \ldots, x_n)^T$.
(3) The mixing process can be represented by a mixed matrix $A \in R^{n\times m}$.

Based on the above assumptions, a simple BSS model can be expressed as:

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_m \end{pmatrix} \tag{1}$$

For convenience, Equation (1) can be expressed as:

$$X = AS \tag{2}$$

The "blindness" of BSS means that the source signal and the mixing matrix are unknown. Therefore, the main task of BSS is to estimate the source signal S and mixing matrix A by only observing the signal X.

The problem of BSS can be divided into three situations according to the relationship of the observed signal number n and the source signal number m.

- If $m < n$, then the problem is overdetermined.
- If $m = n$, then the problem is positive-definite.
- If $m > n$, then the problem is underdetermined.

Different approaches are used in this study to address the BSS problems in different situations. However, before dealing with such BSS problem, its type needs to be determined using signal number estimation.

2.2. Signal number estimation

In the BSS problem, the accuracy of signal number estimation is directly related to the separation performance. Achieving a precise signal number estimation for BSS is a pre-requisite. In this study, the weight potential function based on single-source point detection is used to estimate the signal number. To realize the signal number estimation, we need to assume that the signal is a sparse signal. A signal with a value that is zero or close to zero at most times and is non-zero at other times is called a sparse signal.

If the source signal is sparse, then it can be expressed as Equation (3) at time t.

$$\begin{cases} s_i(t) \neq 0 \\ s_j(t) = 0 \end{cases} j \neq i \; j = (1, 2, \ldots, m) \tag{3}$$

According to Equations (1) and (3), the relationship at time t is expressed as follows:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_i \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} \begin{pmatrix} 0 \\ \vdots \\ s_i \\ \vdots \\ 0 \end{pmatrix} \tag{4}$$

If $A = \{a_1, a_2, \ldots, a_m\}$, then Equation (3) is simplified into Equation (5).

$$\begin{cases} x_j = 0, j \neq i \\ x_i = a_i s_i \end{cases} \tag{5}$$

Through the above analysis, we conclude that the observed signal has linear clustering characteristics when the source signal is sparse enough. The clustering direction is the same as the column vector of the mixed matrix. The center number of clustering is exactly equal to the column number of the mixed matrix.

However, most signals are not sparse enough, which is actually a problem. Some tricks are also used to sparse the signal before signal number estimation. In this study, fast Fourier transform (FFT) and single-source detection are used to obtain the sparse signals. Single-source detection in the transform domain can filter the samples that are not sufficiently sparse. After FFT and single-source point detection, the weight potential function is utilized to estimate the signal number. We assume that the set of single-source samples is defined as $D = \{x_1, x_2, .., x_P\}$, where P represents the number of single-source samples.

In this study, the weighted potential function is defined as follows:

$$J(z) = \sum_{j=1}^{P} \left[\operatorname{sinc}\left(\frac{\cos\left(\theta_{\widehat{x_j,z}}\right)^{-1}}{\sigma^2} \right) \right]^{\gamma} \tag{6}$$

where σ represents the normalized item, z represents the cluster center, $\widehat{x}_j$ represents the normalized observation signal (where $\widehat{x}_j = \frac{x_j}{\|x_j\|}$), $\theta_{x_j,z}$ represents the angle between x_j and clustering centerz, and $\gamma > 0$ represents the weighted potential function adjustment factor.

Sinc function is an oscillation curve with a large main lobe and two decreasing side lobes. When $\cos\left(\theta_{\widehat{x_j,z}}\right) = 1$, the sinc function gets the maximum value. When $\theta_{x_j,z}$ is small, the value of the sinc function is larger, and when $\theta_{x_j,z}$ is larger, the sinc function will be close to zero. Therefore, the signal number will be effectively estimated by the weight potential function.

2.3. Independent component analysis

After the signal number estimation, the type of BSS problem (positive-definite, overdetermined, or underdetermined) is con-

firmed. The positive-definite case is usually the simplest case, and it has been widely and comprehensively studied [24]. In this part, ICA is used to deal with the BSS problem in the positive-definite case. The ICA algorithm is usually based on the following assumptions:

- Source signals are statistically independent of one another.
- The distribution function of the source signal is non-Gaussian.
- The source signal number is equal to the observed signal number.

According to the central limit theorem, the sum of independent random variable is closer to the Gaussian distribution than any original random variable. Therefore, according to Equation (1), the observed signal X, which can be defined as the source signal sum, is closer to the Gaussian distribution than any source signal s_i.

To obtain the source signal, we define a linear combination of such signal as follows:

$$y = w^T X = \sum_{i=1}^{n} w_i x_i = w^T AS \tag{7}$$

where y is the linear combination of s_i. Thus, y is closer to the Gaussian distribution than any s_i. The Gaussian characteristic of y is the weakest when w is chosen such that $y = s_i$ ($w^T A$ is the unit vector). Therefore, the source signal is recovered when the Gaussian characteristics of y are minimized. To minimize the Gaussian characteristic of the signal, we need to select an appropriate Gaussian characteristic metric. In this study, negative entropy is introduced for the Gaussian characteristic metric. The definition of negative entropy $J(y)$ is expressed as follows:

$$J(y) = H(y_{gauss}) - H(y) \tag{8}$$

where y_{gauss} is the Gaussian random variable with the same variance as$y$, and $H(y)$ denotes the differential entropy, which is defined as follows:

$$H(y) = -\int P(y)\log(P(y)) \tag{9}$$

where $P(y)$ is the probability density function of$y$. In practice, the negative entropy can be simplified as Equation (10).

$$J(y) = k_1(E\{G_1(y)\})^2 + k_2(E\{G_2(y)\} - E\{G_2(v)\})^2 \tag{10}$$

where k_1 and k_2 are constants, v is a standardized Gaussian random variable with zero mean and unit variance, y is a standardized random variable, and G is an arbitrary non-quadratic function. A set of G functions is shown as follows:

$$\begin{aligned} G_1(y) &= \tfrac{1}{a_1}\log(\cosh(a_1 y)) \\ G_2(y) &= -\exp\left(-y^2/2\right) \end{aligned} \tag{11}$$

where a_1 is usually equal to 1. In this study, a fixed-point iterative algorithm is used to find the non-Gaussian maximum of $w^T X$. In addition, negative entropy is used as a non-Gaussian metric for FastICA. The iteration formula of the FastICA algorithm is shown as follows:

$$w(k+1) = E\left\{xg\left(w(k)^T x\right)\right\} - E\left\{g'\left(w(k)^T x\right)\right\}w(k) \tag{12}$$

where $g(y) = y\exp(-y^2/2)$, and g' is a derivative of g. A key step to whiten the observed signal must be performed before the FastICA algorithm is implemented. The algorithm converges quickly through the whitening process. Centralization is the first step in whitening and leads to the mean of the observed signal becoming zero. The variance of the observed signal then becomes a unit in the normalized process. The whitening process can be expressed as follows:

$$\begin{aligned} E(X) &= \mathbf{0} \\ E\left(XX^T\right) &= I \end{aligned} \tag{13}$$

where I is the unit matrix, and $\mathbf{0}$ is the zero vector. The main steps of FastICA are shown in Algorithm 1.

Algorithm 1 FastICA algorithm main steps

(1) Observation vectors are whitened;
(2) The vector w is initialized;
(3) The iteration is done,
$w(k+1) = E\left\{xg\left(w(k)^T x\right)\right\} - E\left\{g'\left(w(k)^T x\right)\right\}w(k)$;
(4) The vector w is normalized, $w = w/\| w \|$;
(5) If the iteration does not converge, return step 3.

2.4. Principal component analysis

In this part, the BSS problem in the overdetermined case is discussed. When observed signals are more than the source signals, the ICA is no longer suitable for this case. The PCA algorithm [25,26] is introduced to reduce the dimension of observed signals so that the conditions of the ICA algorithm are satisfied.

A linear combination of observed signals is considered as follows:

$$z_1 = b_1^T X \tag{14}$$

where b_1 is the weight vector. If the vector b_1 is adjusted to maximize the variance of z_1, then z_1 is defined as the first principal component of the observed signal X. Given that the variance of the z_1 is affected by the vector b_1, the norm of b_1 needs to be bound to 1. Hence PCA indicators are maximized as follows:

$$\begin{aligned} J(b_1) &= E\{z_1^2\} = b_1^T C_x b_1 \\ &\text{st}: \| b_1 \| = 1 \end{aligned} \tag{15}$$

where $\| \cdot \|$ is the Euclidean norm, and C_x is the covariance matrix of x. According to linear algebra theory, the PCA solution is a standardized eigenvector of C_x. If the eigenvector $e_1, \ldots, e_n$ corresponds to the eigenvalue $d_1, \ldots, d_n$ one by one and $d_1 \geqslant d_2 \geqslant \cdots \geqslant d_n$, then the first principal component can be expressed as follows:

$$z_1 = e_1^T X \tag{16}$$

The result in Equation (16) is generalized to m principal components.

$$Z = BX = (e_1, e_2, \ldots, e_m)^T X \tag{17}$$

When the dimension of Z is equal to the dimension of the source signal, the condition of ICA is satisfied. In this case, the ICA algorithm can separate the source signal from the principal component.

2.5. Variational modal decomposition

Underdetermined BSS may be the most challenging BSS task. In this part, VMD [27] is used to construct virtual channels to make the ICA suitable for underdetermined case. The virtual channel is used to supplement the dimensions of the observed signal so that ICA conditions are satisfied. VMD decomposes the observed signal into K intrinsic mode functions (IMFs) with different center frequencies [28]. IMF is defined as an AM-FM signal, as shown in Equation (18).

$$u_i(t) = A_i(t)\cos(\phi_i(t)) \tag{18}$$

where i is the ith IMF, $A_k(t)$ is the amplitude modulation function, and $\phi_i(t)$ is the phase modulation function. The analytic signal is obtained from the IMF through the Hilbert transform as Equation (19).

$$\left(\delta(t) + \frac{j}{\pi t}\right)u_i(t) \tag{19}$$

where $\delta(t)$ is the impulse function. Next, the central frequency of the analytic signal is migrated as follows:

$$\left[\left(\delta(t) + \frac{j}{\pi t}\right)u_i(t)\right]e^{-jw_i t} \tag{20}$$

Finally, each component bandwidth is measured by the square of the $L2$ norm. By minimizing the sum of each component bandwidth, the constrained variational model is established as Equation (21).

$$\min_{\{u_i\},\{w_i\}} \left\{\sum_i \| \partial_t\left[\left(\delta(t) + \tfrac{j}{\pi t}\right)u_i(t)\right]e^{-jw_i t} \|_2^2\right\} \quad s.t.\ \sum u_i(t) = f \tag{21}$$

where $\{u_i\}$ is the set of IMFs, $\{ w_i\}$ is the set of IMF central frequencies, and f is the analytic signal.

Based on the model in Equation (21), the constrained variational problem can be transformed into an unconstrained variational problem by introducing penalty factor α and Lagrange multiplier λ as follows:

$$\begin{aligned} L(\{u_i\},\{\omega_i\},\lambda) = \alpha\sum \| \partial_t\left[\left(\delta(t) + \frac{j}{\pi t}\right) * u_k(t)\right]e^{-jw_i t} \|_2^2 \\ + \| f(t) - \sum \omega_i(t) \|_2^2 \\ + \left\langle \lambda(t), f(t) - \sum u_k(t)\right\rangle \end{aligned} \tag{22}$$

The alternate direction method of multipliers (ADMM) is used to solve Equation (22). The signal is decomposed adaptively through constant updating. With ADMM, the problem is equivalent to Equation (23).

$$u_i^{n+1} = \arg\min\left\{\alpha\| \partial_t\left[\left(\delta(t) + \frac{j}{\pi t}\right)u_k(t)\right]e^{-jw_i t} \|_2^2 + \| f(t) - \sum u_i(t) + \frac{\lambda(t)}{2} \|_2^2\right\} \tag{23}$$

The problem is converted into the frequency domain as Equation (24) by using the Parseval/Plancherel Fourier isometric transform.

$$\hat{u}_i^{n+1} = \arg\min\left\{\alpha\| j\omega[(1 + sgn(\omega + \omega_i))\hat{u}_i(\omega + \omega_i)] \|_2^2 + \| \hat{f}(\omega) - \sum_l \hat{u}_l(\omega) + \frac{\hat{\lambda}(\omega)}{2} \|_2^2\right\} \tag{24}$$

Finally, the VMD main steps are shown in Algorithm 2.

Algorithm 2 VMD algorithm main steps

(1) $u_i \omega_i\ \lambda\ n$ is initialized;

(2) u_i iteration:$\hat{u}_i^{n+1}(\omega) = \frac{\hat{f}(w) - \sum_{l\neq i}\hat{u}_i(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega - \omega_i)^2}$;

(3) ω_i iteration:$\omega_i^{n+1} = \frac{\int_0^\infty \omega|\hat{u}_i(\omega)|^2 d\omega}{\int_0^\infty |\hat{u}_i(\omega)|^2 d\omega}$;

(4) λ iteration:$\hat{\lambda}^{n+1}(\omega) = \hat{\lambda}^n(\omega) + \tau\left(\hat{f}(w) - \sum_l \hat{u}_l^{n+1}(\omega)\right)$;

(5) If $\sum_l \| \hat{u}_l^{n+1} - \hat{u}_l^n \|_2^2 / \| \hat{u}_l^n \|_2^2 < \varepsilon$, end the iteration, otherwise return to the step

3. Experiment

In this section, we verify the performance of the proposed method. This section is mainly divided into three parts. In the first part, we will briefly introduce the experimental signals. In the second part, we verify the validity of the source number estimation algorithm. In the third part, we detect the UAV sounds with mixed signal in overdetermined, positive-definite, and underdetermined cases. All experiments are conducted using MATLAB.

3.1. Signal description

The sound signals in the experiment include two parts: UAV sound and interference sounds. The UAV sounds come from DJI spark, DJI Phantom 4 Pro and AMOVLAB Z410, as shown in Fig. 3. All UAV sounds are recorded by skc-mp40 microphone sensor when the UAVs hover at a constant height of 2 m. The sound length of each UAV is 30 s, and the sampling rate is 48kS/s. We divide the UAV sound into test set and training set, and their ratio is 1:4. More specifically, the sound is divided into 5 s data blocks, in which one second sound is randomly selected and put into the test set, and the remaining sound is put into the training set.

Fig. 4 shows the time domain and time–frequency domain information of three UAV sounds. It needs to be pointed out that the energy of UAV sound is concentrated in the frequency band of 0–3 kHz, but some energy is also observed in the high frequency part.

The interference sounds come from the open source sound library UrbanSound [29], which contains a variety of urban environment sounds. We select gun, jackhammer and siren as interference sounds. These signals include abrupt sound (gun), discontinuous sound (jackhammer), and periodic sound (siren); the spectrum of these sounds also differ. The time–frequency characteristics of the interference signals are shown in Fig. 5. According to Figs. 4 and 5, UAV and interference signals overlap in frequency and time domains. Therefore, the traditional filtering method cannot eliminate the interference mixed with UAV sound. In this case, BSS can be used to separate UAV sound.

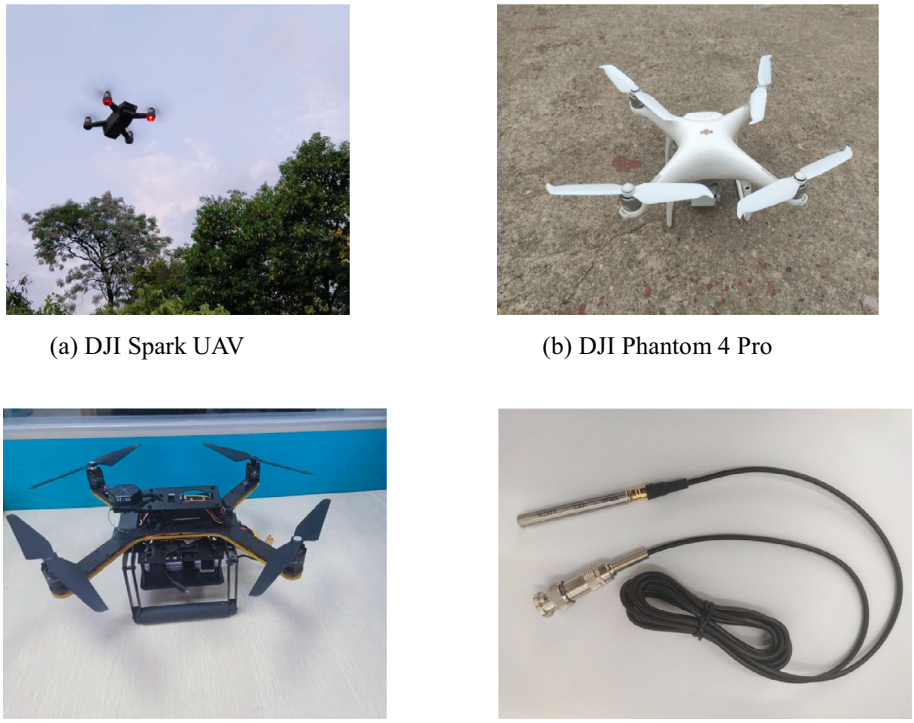
3.2. Signal number estimation

In the signal number estimation experiment, the 2×2 matrix is used to mix the UAV sound (Spark UAV is selected here for the experiment) and jackhammer sound. Directional clustering characteristics are not found in the time domain of the observed signal, as shown in Fig. 6(a). Although the directional clustering feature of the observed signal is found in the frequency domain, the directional clustering feature near the origin is covered by interference and noise due to the lack of sparsity, as shown in Fig. 6(b). Therefore, the single source detection method is used to improve sparsity, as shown in Fig. 6(c). The directional clustering in Fig. 6c becomes more obvious than that in Fig. 6(b).

The calculation process of three and four sources is the same as the aforementioned process; thus, it is not repeated here. As shown in Table 1, different mixing matrices are used to mix different signal numbers.

Fig. 7 shows the results of signal number estimation. Fig. 7(a) shows the two-source number estimation, in which two peaks can be seen and correspond to the source signal number (two sources). Fig. 7(b) and (c) respectively have three and four peaks, which are consistent with their corresponding source numbers.

Fig. 8 shows a test used to evaluate the accuracy of source number estimation. The mix matrices are generated by random functions in the test. The accuracy is obtained by calculating the



(a) DJI Spark UAV (b) DJI Phantom 4 Pro

(c) AMOVLAB Z410 (d) SKC-MP40 microphone

Fig. 3. Experimental equipment.

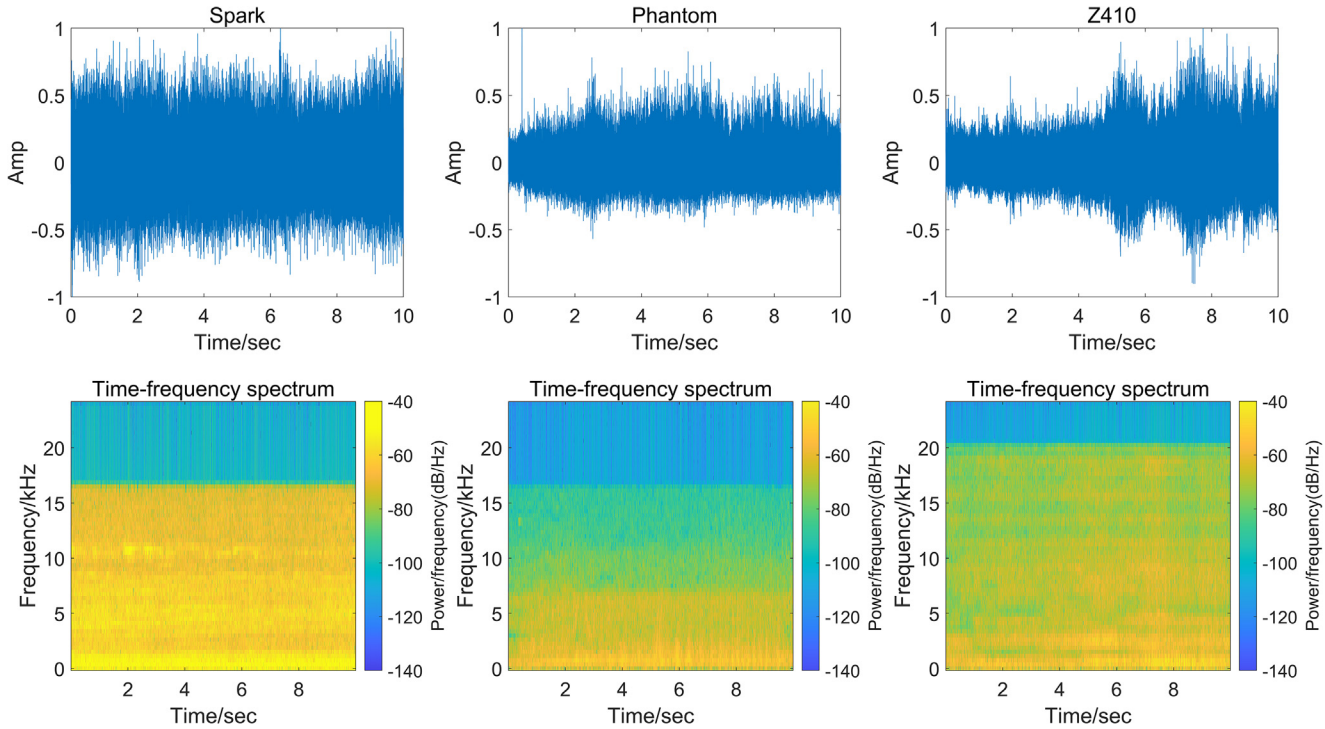


Fig. 4. Tim-frequency characteristics of UAV sounds.

percentage of the number of correct identifications in the total number of tests.

3.3. Sound separation and UAV detection

Sound separation and UAV detection are discussed in this part. This section consists of three experiments. In the first experiment, we consider the performance of this algorithm in different machine learning models. In the second experiment, different sound features are considered; In the third experiment, the performance of the algorithm in different UAVs is assessed. In the three experiments, we use random matrix to mix sound signals. In order to avoid the influence of extreme values, we set the final result as the average result of 50 experiments. For comparison, the observa-

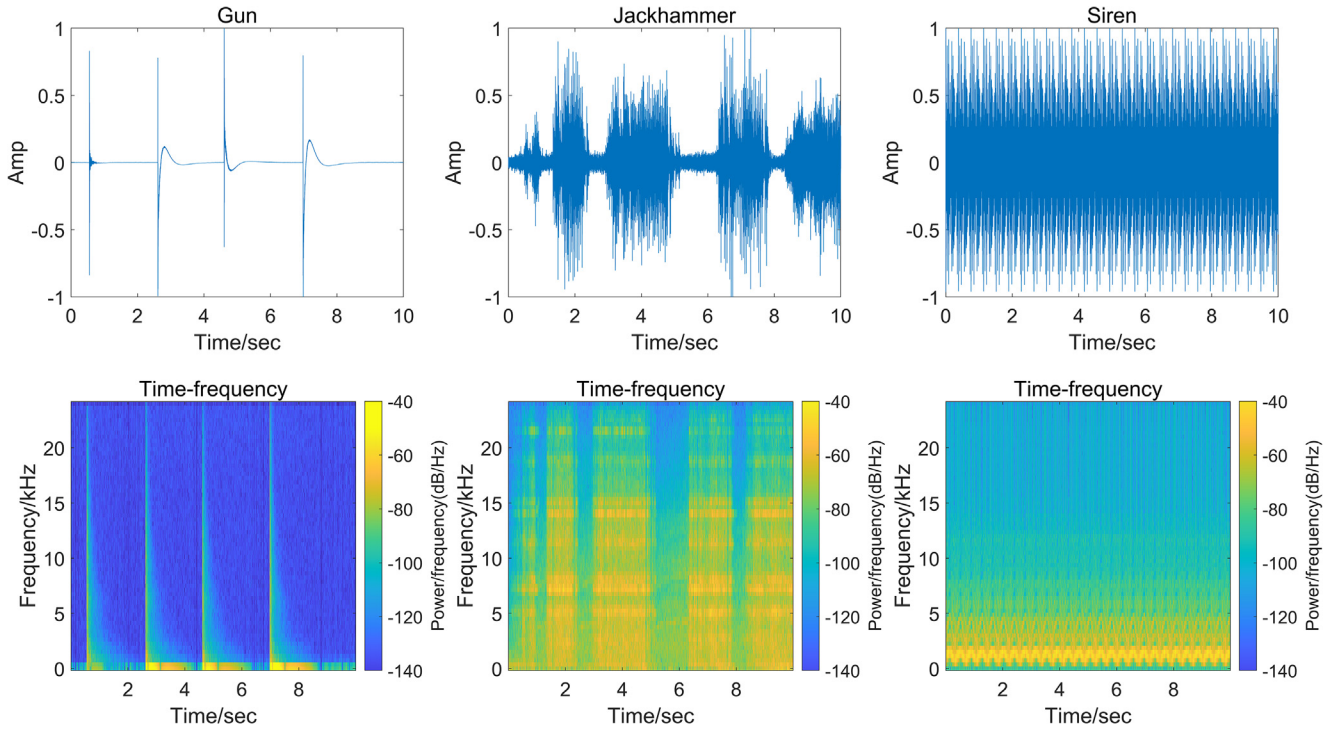


Fig. 5. Time-frequency characteristic of interference sounds.

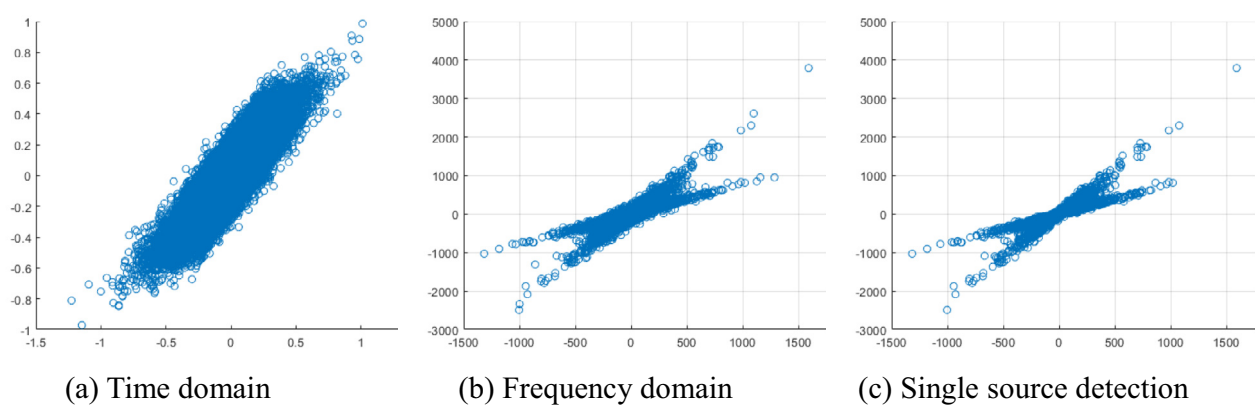


Fig. 6. Data processing results.

Table 1
Mix matrix.

Two-source	Three-source	Four-source
$\begin{bmatrix} 0.28 & 0.97 \\ 0.65 & 0.74 \end{bmatrix}$	$\begin{bmatrix} 0.22 & 0.74 & 0.33 \\ 0.32 & 0.74 & 0.83 \end{bmatrix}$	$\begin{bmatrix} 0.73 & 0.93 & 0.78 & 0.39 \\ 006 & 0.98 & 0.51 & 0.13 \end{bmatrix}$

tion signal (mixed signal) and the signal filtered by a filter (filter signal) are used to identify the UAV in three experiments. Accuracy (ACC) is used to measure the performance of the algorithm in the experiments. ACC is defined as follows:

$$\mathrm{ACC} = \frac{TP + \mathrm{TN}}{TP + TN + FP + FN} \times 100\% \tag{25}$$

In order to compare the performance of blind source separation algorithm and traditional filter algorithm on anti-interference performance, this paper analyzes the frequency domain energy distribution of UAV sound and interference sounds, and the results are shown in Fig. 9. It can be seen from Fig. 9 that 50 % of the UAV sound energy is distributed at 0–3 kHz, and the interference sounds such as siren and gun also have strong energy at 0–3 kHz. According to the sound energy distribution of UAV, we choose 0–3 kHz low-pass filter to compare with BSS method.

In the first experiment, we mainly consider the stability of the proposed algorithm in different machine learning algorithms. The experimental results are shown in Fig. 10. The ACC of the observed signal decreases gradually from over determined to under determined because of the gradual increase of the interference source number. The ACC of the filter signal is better than that of observation signal. However, two problems arise for the filter signal: the ACC decays greatly with the increase of interference, and the recognition rate of the filter signal by the KNN and decision tree algorithms is lower than that of the observation signal in the positive-definite case. Hence, the filtering method is not a robust

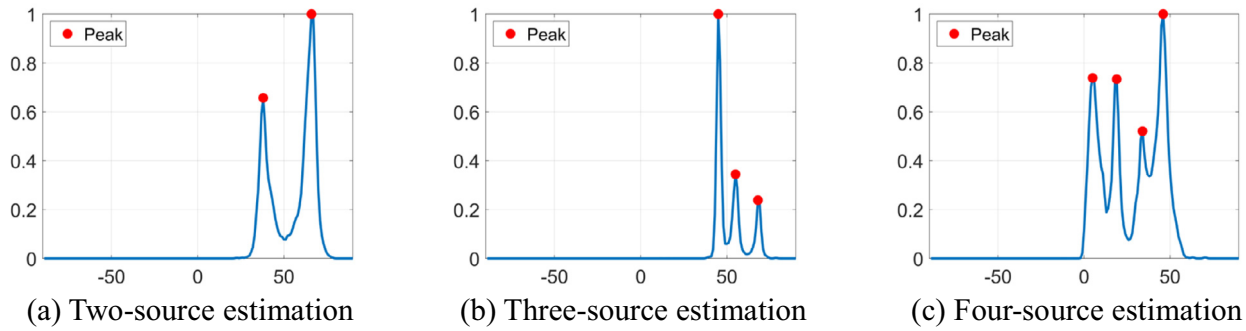


Fig. 7. Source number estimation.

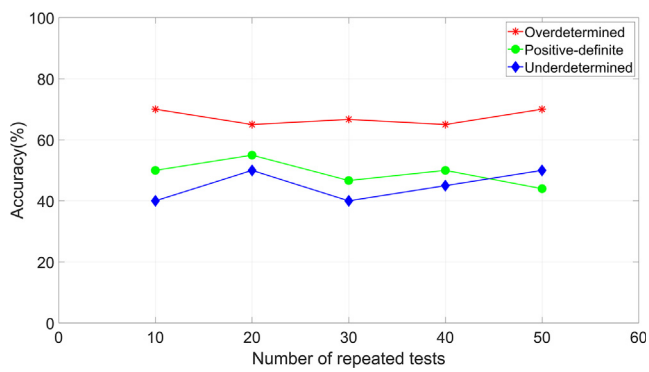


Fig. 8. Accuracy of source number estimation.

anti-interference strategy, at least on this issue. Although the ACC of separated signals also has a downward trend, the recognition rate remains within a certain range. Among different machine learning algorithms, SVM and KNN methods are obviously better than decision tree methods for observation signals and filtered signals. No matter which machine learning method is used, The ACC of our scheme herein is more than 90 %, and its performance is obviously better than that of observation signal and filtering signal.

In the second experiment, the algorithm performance is assessed using different sound features, and the results are shown in Fig. 11. MFCC and GTCC features, which are two very common features in the field of sound recognition, are used in this experiment. The experimental results show that MFCC is slightly better than GTCC in the overdetermined and underdetermined cases. In the positive-definite case, the result of MFCC is obviously better than that of GTCC. Regardless of the feature, the ACC of the algorithm in this work can reach more than 90 %. The ACC of the two features using the mixed and filter signals gradually decreases to about 50 % with the increase of interference.

In experiment 3, the performance of the proposed algorithm in different UAVs is tested, and the results are shown in Fig. 12. The experimental results show that with more interference sources, the ACC of the mixed signals in the three UAVs decreases gradually and is less than 60 % in the underdetermined case. The proposed

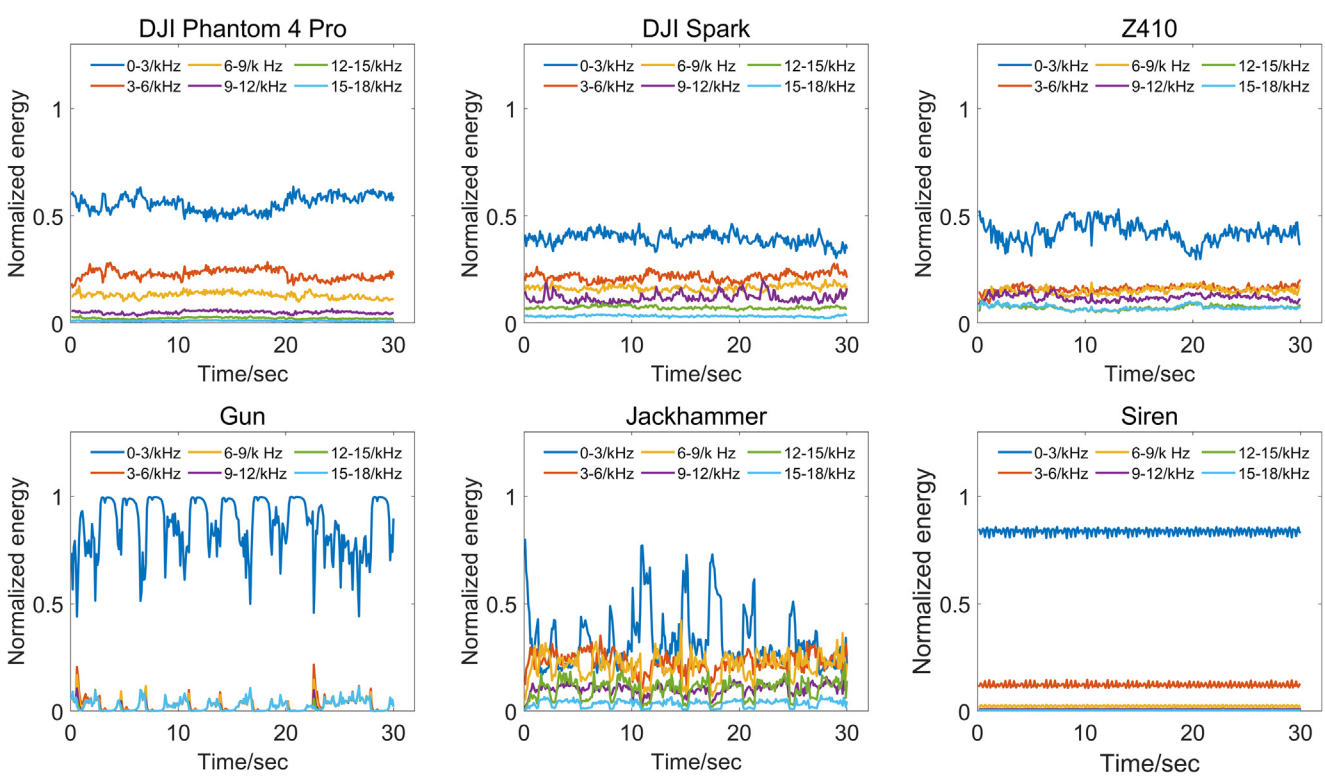


Fig. 9. Signal energy distribution.

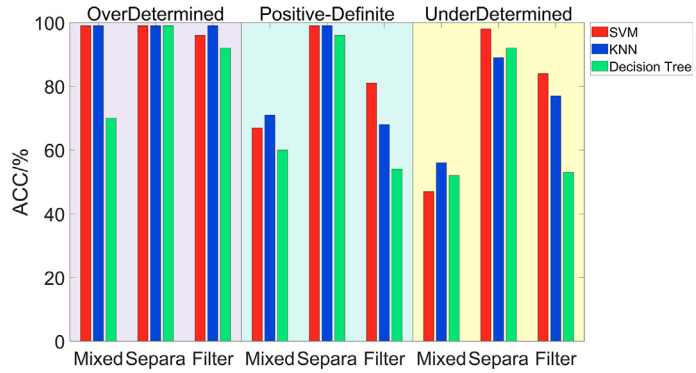


Fig. 10. UAV detection through different ML methods.

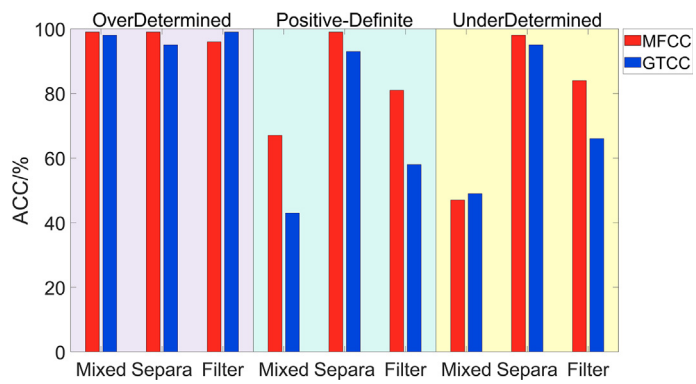


Fig. 11. UAV detection in different sound Features.

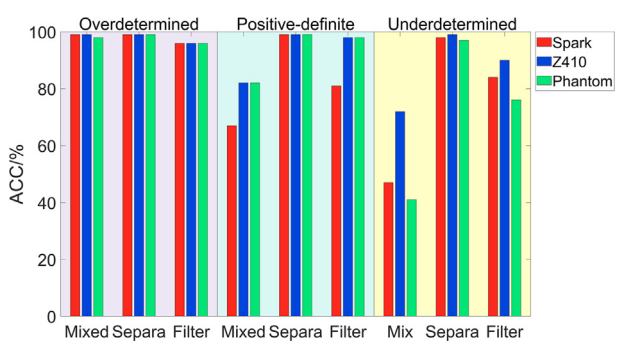


Fig. 12. UAV detection in different UAVs.

method shows good performance in different UAVs, and the ACC is always more than 90 %. Although the ACC of the filtered signal is increased relative to that of the mixed signal, its performance is always inferior to that of the proposed method.

4. Conclusion

Acoustic anti-UAV technology is developing rapidly because of active development of UAVs. However, existing schemes rarely involve UAV detection with strong interference in overdetermined or underdetermined cases. Thus, this paper study discusses the UAV detection methods based on the basis of the BSS framework. This scheme unifies multiple BSS methods into our framework and realizes UAV detection in different cases by using sound signals with strong interference. We estimate the signal number, determine the type of BSS problems, and then realize UAV detection by using machine learning. The experiments using different machine learning methods and sound features are performed, and their results show that the performance of our scheme is better than the mixed signal and traditional filtering. The test of multiple UAVs shows that the performance of our scheme achieves a good recognition effect for different UAVs. Compared with the traditional filter scheme, the ACC of UAV sound detection method based on BSS framework is more than 90 %. In the future, we will consider improving our scheme to boost the running speed of the algorithm. This scheme will be deployed to embedded devices to test its performance in our plan.

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CRediT authorship contribution statement

Wenshuai Wang: Conceptualization, Methodology, Software, Writing – original draft. **Kuangang Fan:** Writing – review & editing, Funding acquisition. **Qinghua Ouyang:** Software. **Ye Yuan:** Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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